

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 13

Date Given: November 21, 2025

Date Due: November 28, 2025
Total Points: 10

1. Find the *moment generating function (mgf)* of χ_n^2 -distribution. What is the domain of definition of this mgf?
2. Consider a random variable X which has the distribution $\mathbf{P}(X = n) = \frac{6}{(\pi n)^2}$ for $n \geq 1$. Show that its mgf is defined only for $t \leq 0$.
3. Show that there is a random variable X , such that, its mgf is only defined for $t = 0$. [**Hint:** You may use the **Problem # 2** above.]
4. Suppose ϕ is a *characteristic function* and f is a pgf. then show that $f \circ \phi$ is a characteristic function.